

EDA

Exploratory data analysis understand dataset structure detect outlier identify missing values before Machine Learning

TYPES Univariate,Bivariate,multivariate.

```
In [2]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset("tips")
print(df.head())
print(df.info())
print(df.columns)
```

```
   total_bill  tip  sex smoker  day  time  size
0      16.99  1.01 Female    No  Sun  Dinner    2
1      10.34  1.66  Male    No  Sun  Dinner    3
2      21.01  3.50  Male    No  Sun  Dinner    3
3      23.68  3.31  Male    No  Sun  Dinner    2
4      24.59  3.61 Female    No  Sun  Dinner    4
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 244 entries, 0 to 243
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	total_bill	244 non-null	float64
1	tip	244 non-null	float64
2	sex	244 non-null	category
3	smoker	244 non-null	category
4	day	244 non-null	category
5	time	244 non-null	category
6	size	244 non-null	int64

```
dtypes: category(4), float64(2), int64(1)
```

```
memory usage: 7.4 KB
```

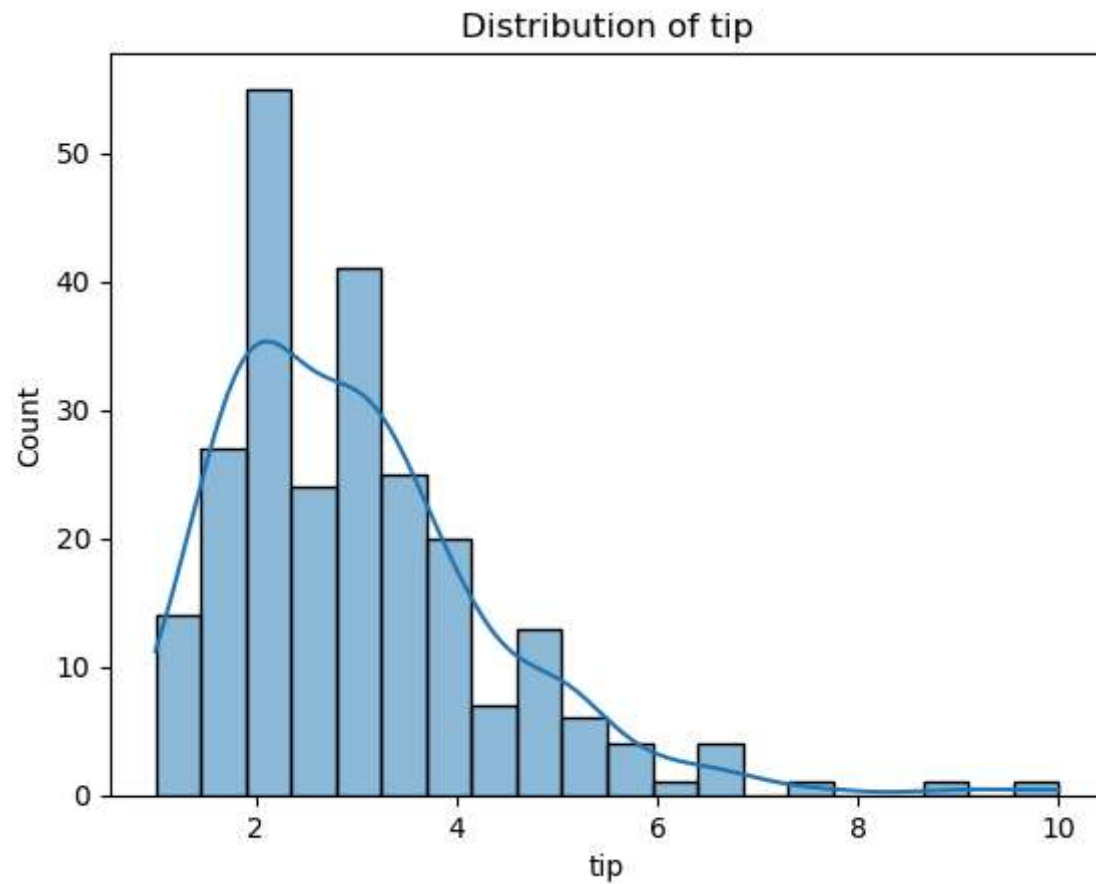
```
None
```

```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

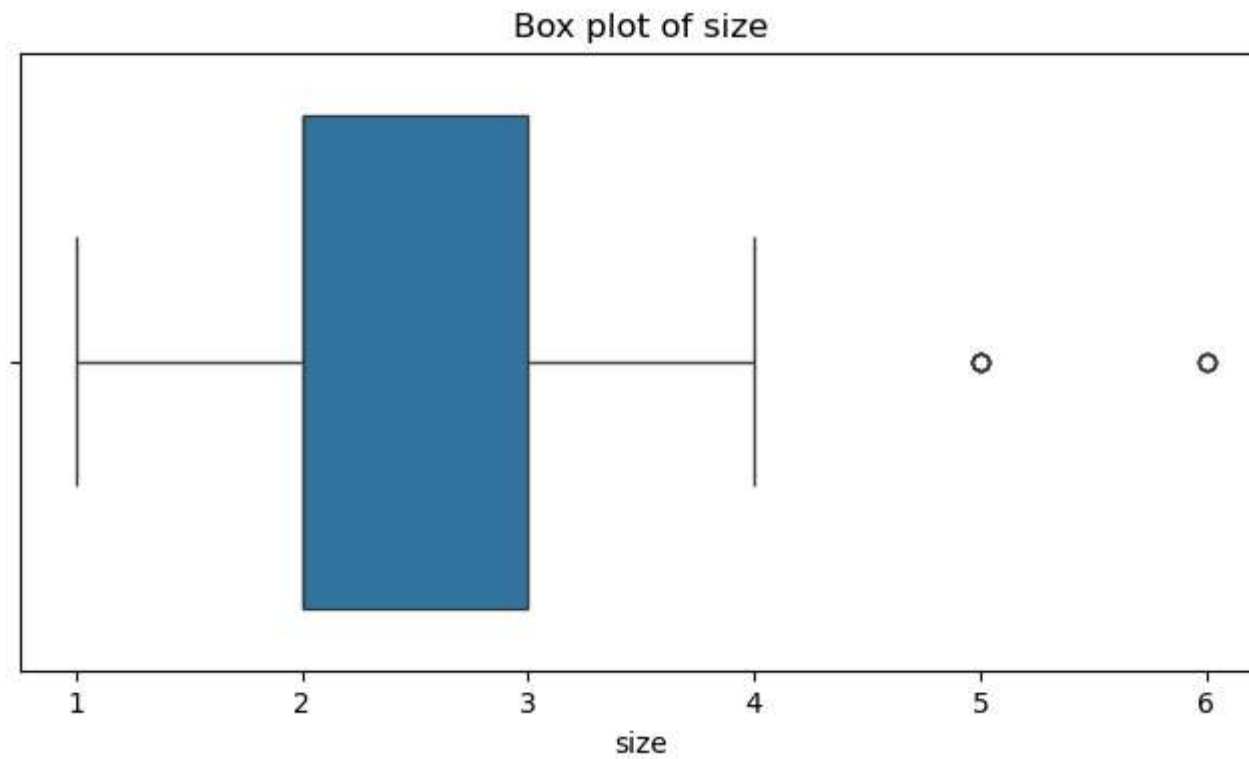
```
In [3]: #1.Descriptive Statistics
print(df['total_bill'].describe())
```

```
#Histogram
sns.histplot(df['tip'],bins=20,kde=True)
plt.title("Distribution of tip")
plt.show()
```

```
count    244.000000
mean      19.785943
std       8.902412
min       3.070000
25%      13.347500
50%      17.795000
75%      24.127500
max       50.810000
Name: total_bill, dtype: float64
```



```
In [4]: #Box Plot
plt.figure(figsize=(8,4))
sns.boxplot(x=df['size'])
plt.title("Box plot of size")
plt.show()
```

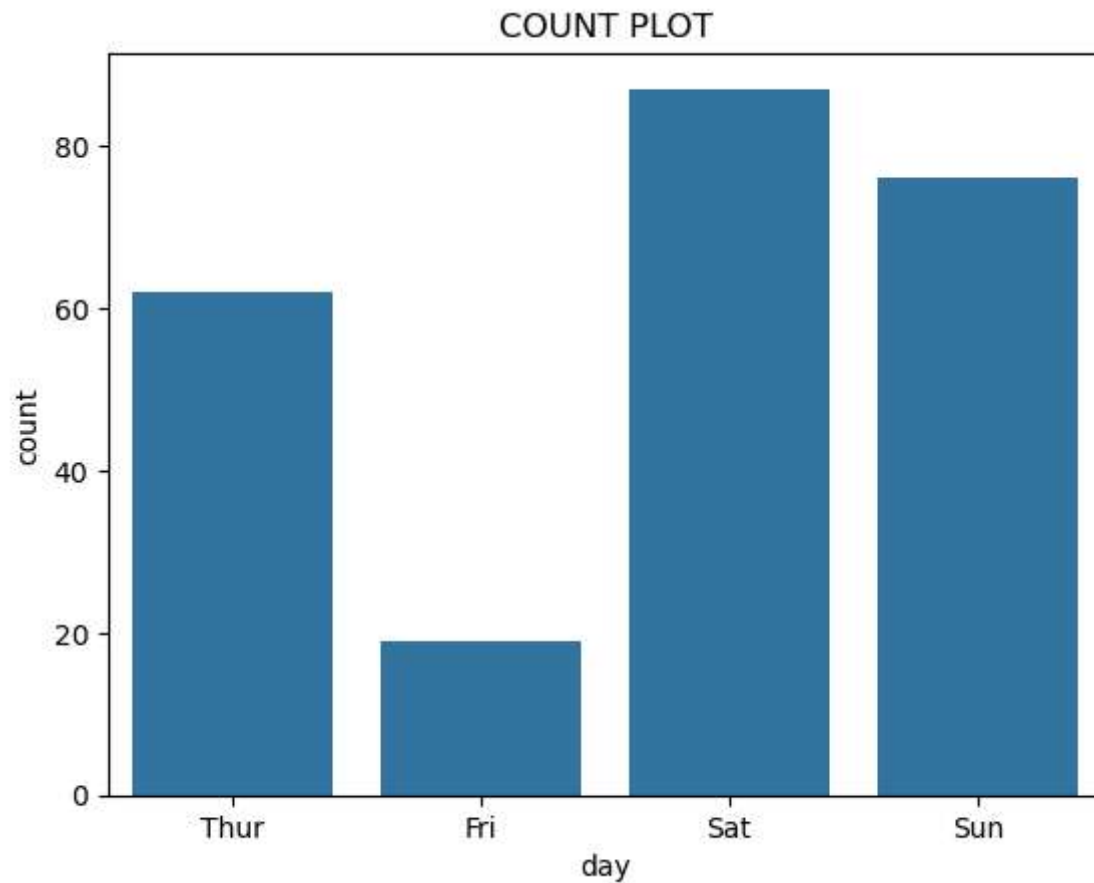


CATEGORICAL VARIABLE DAY

```
In [5]: #VALUE COUNTS
print(df['day'].value_counts())
```

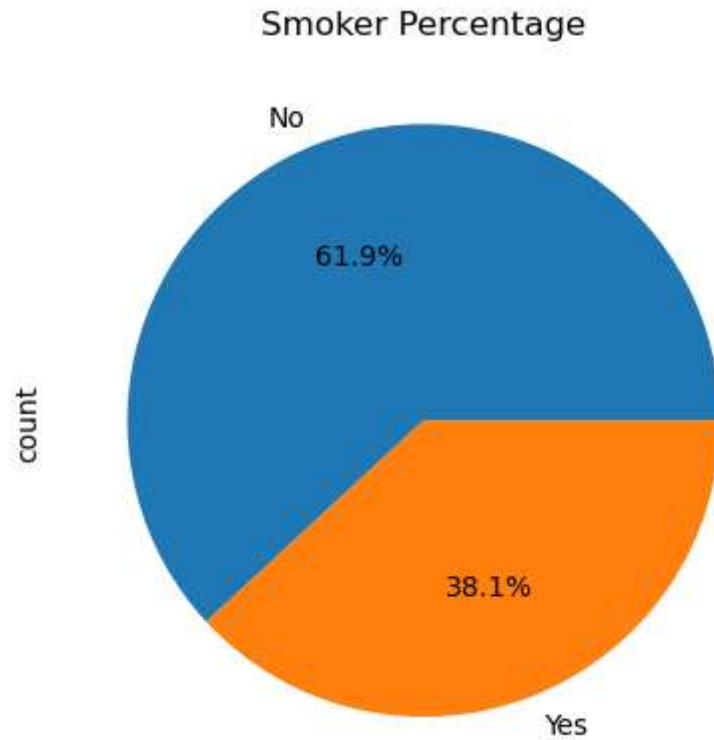
```
day
Sat      87
Sun      76
Thur     62
Fri      19
Name: count, dtype: int64
```

```
In [6]: ## Count plot
sns.countplot(x=df['day'],data=df)
plt.title("COUNT PLOT")
plt.show()
```



```
In [7]: #PIE CHART
df['smoker'].value_counts().plot(kind='pie', autopct='%1.1f%%')
```

```
plt.title("Smoker Percentage")
#plt.ylabel("")
plt.show()
```



```
In [8]: #Outerlier Detection using IQR
Q1=df['total_bill'].quantile(0.25)
Q3=df['total_bill'].quantile(0.75)
IQR = Q3-Q1

lower=Q1-1.5*IQR
upper=Q3+1.5*IQR

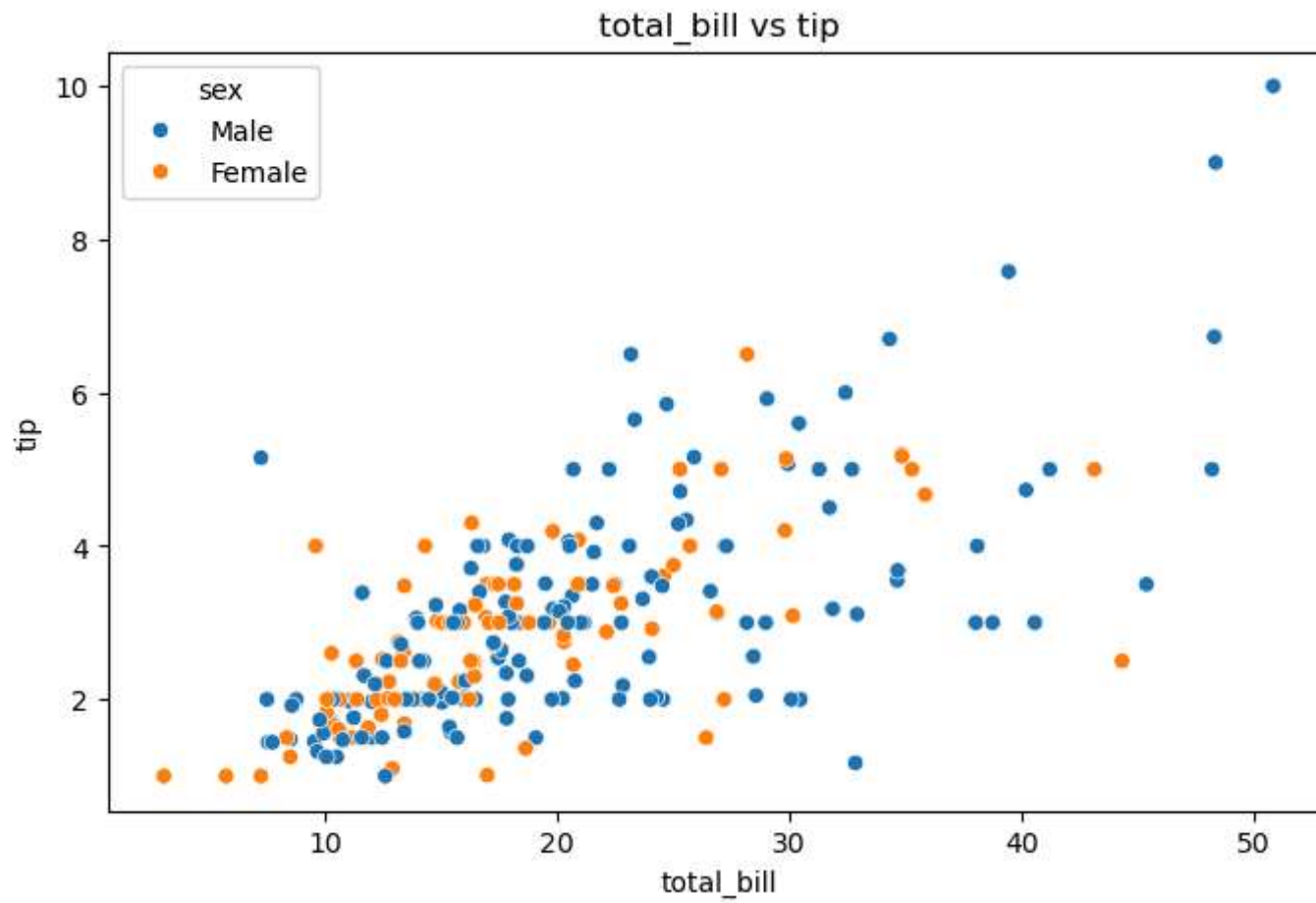
outliers=df[
    (df['total_bill']<lower) | (df['total_bill']>upper)]
print("Outliers:",len(outliers))
```

Outliers: 9

BIVARIATE

types numeric - numeric - Scatterplot category - numeric - bar chart category - category - grouped statistic/cross tap

```
In [9]: #Numeric vs Numeric - total_bill VS tip
#Scatter Plot
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='total_bill',
    y='tip',
    hue='sex',
    data=df
)
plt.title("total_bill vs tip")
plt.show()
```

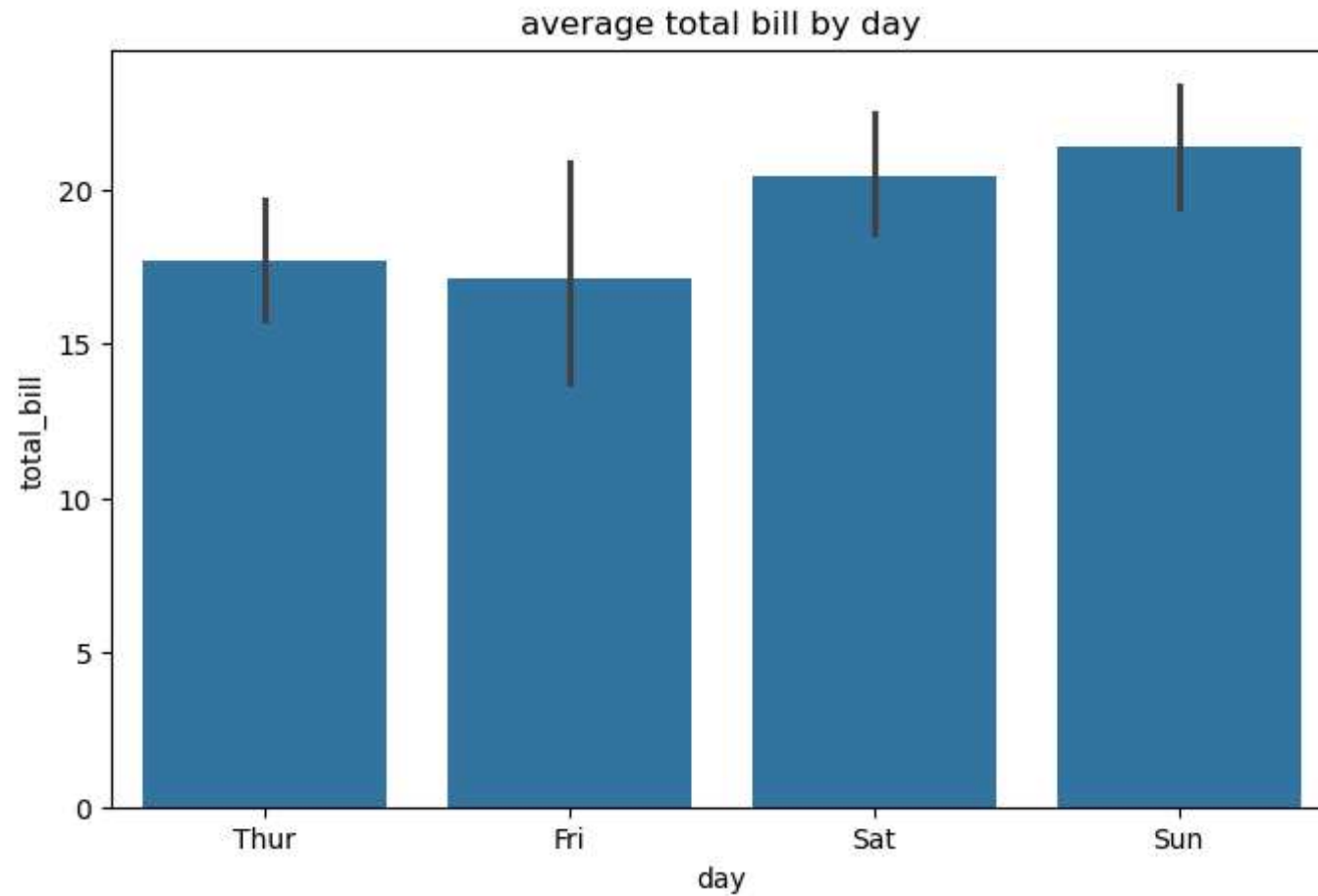


```
In [10]: #CORRELATION
#PEARSON
corr=df['total_bill'].corr(df['tip'])
print(corr)
```

0.6757341092113641

```
In [11]: #CATEGORICAL VS NUMERICAL
#DAY VS TOTAL BILL
plt.figure(figsize=(8,5))
sns.barplot(
    x='day',
    y='total_bill',
    data=df
```

```
)  
plt.title("average total bill by day")  
plt.show()
```



```
In [12]: #GROUPED STATISTIC  
#MEAN BILL BY DAY  
print(  
    df.groupby('day', observed=False)['total_bill'].mean()  
)  
  
#MULTIPLE STATISTICS  
print(  

```

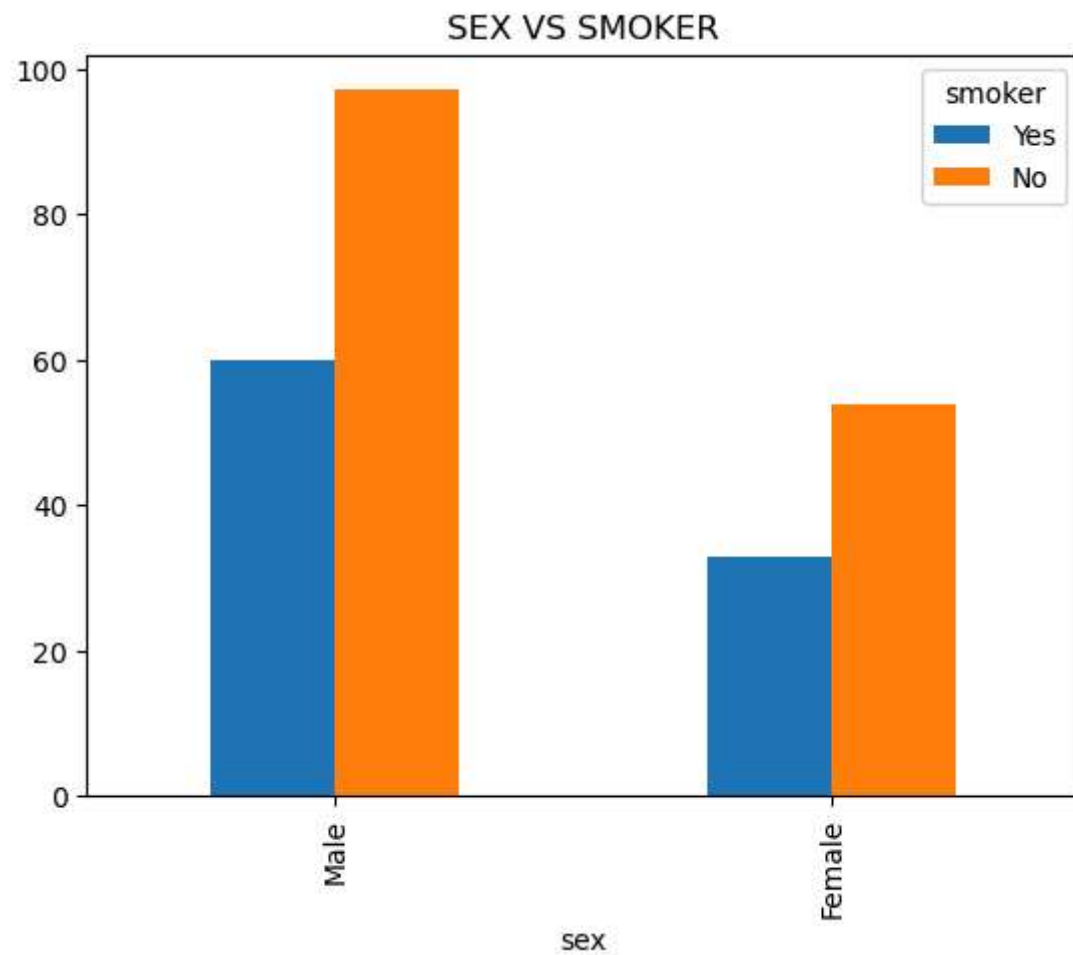


```
df.groupby('day',observed=False)['total_bill'].agg(['mean','median','min','max'])
```

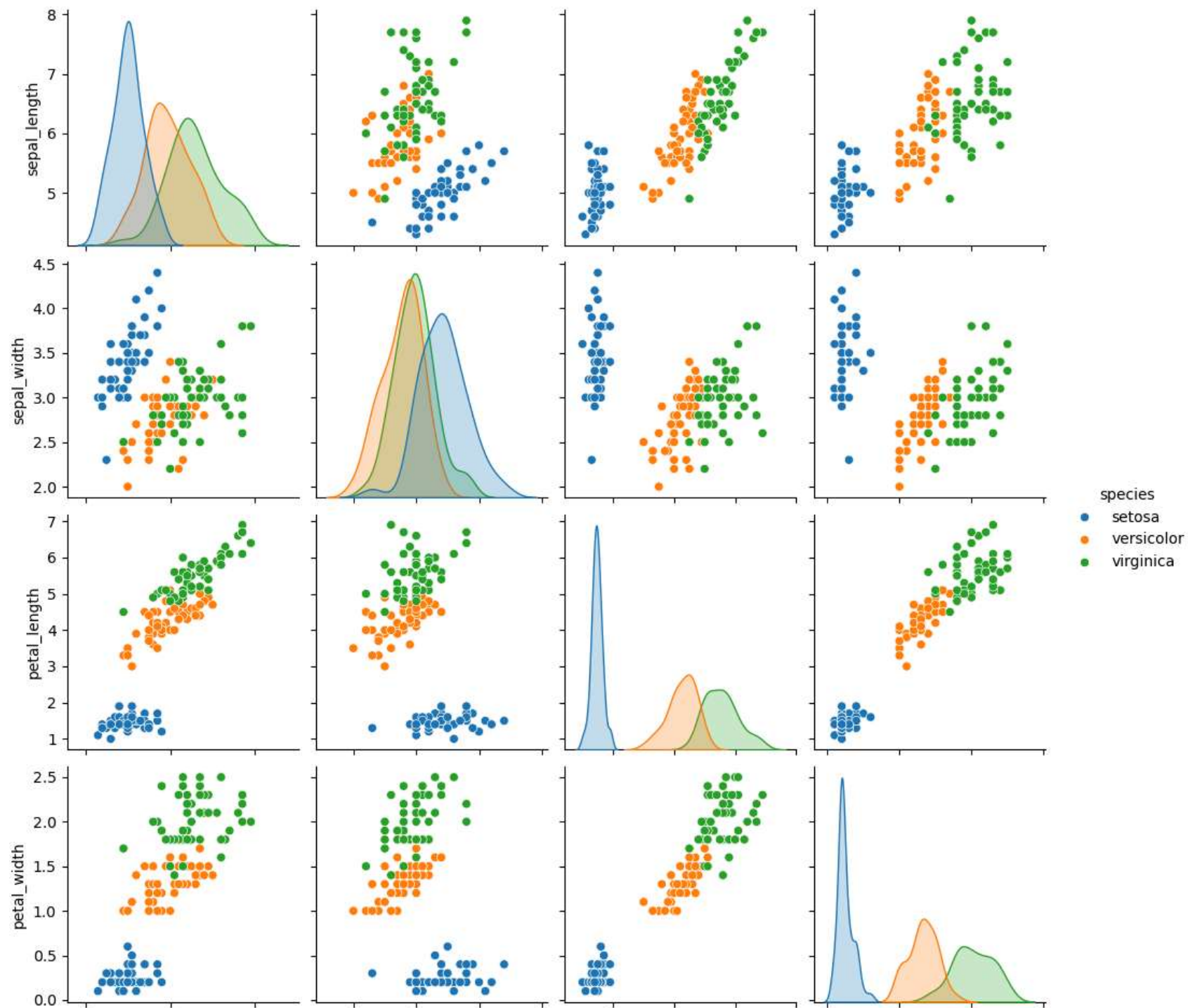
```
day
Thur    17.682742
Fri     17.151579
Sat     20.441379
Sun     21.410000
Name: total_bill, dtype: float64
      mean  median   min   max
day
Thur  17.682742   16.20  7.51  43.11
Fri   17.151579   15.38  5.75  40.17
Sat   20.441379   18.24  3.07  50.81
Sun   21.410000   19.63  7.25  48.17
```

```
In [13]: #CATEGORY VS CATEGORY
#CROSS TAB - (pandas library)- Relationship between two categorical variables.

#BAR CHART
pd.crosstab(
    df['sex'],
    df['smoker']
).plot(kind='bar')
plt.title("SEX VS SMOKER")
plt.show()
```



```
In [14]: #MULTIVARIATE ANALYSIS
#Pair plot
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset("iris")
sns.pairplot(
    df,
    hue='species'
)
plt.show()
```



4 6 8 2 3 4 5 2 4 6 8 0 1 2 3
sepal_length sepal_width petal_length petal_width

```
In [15]: print(df.columns)
```

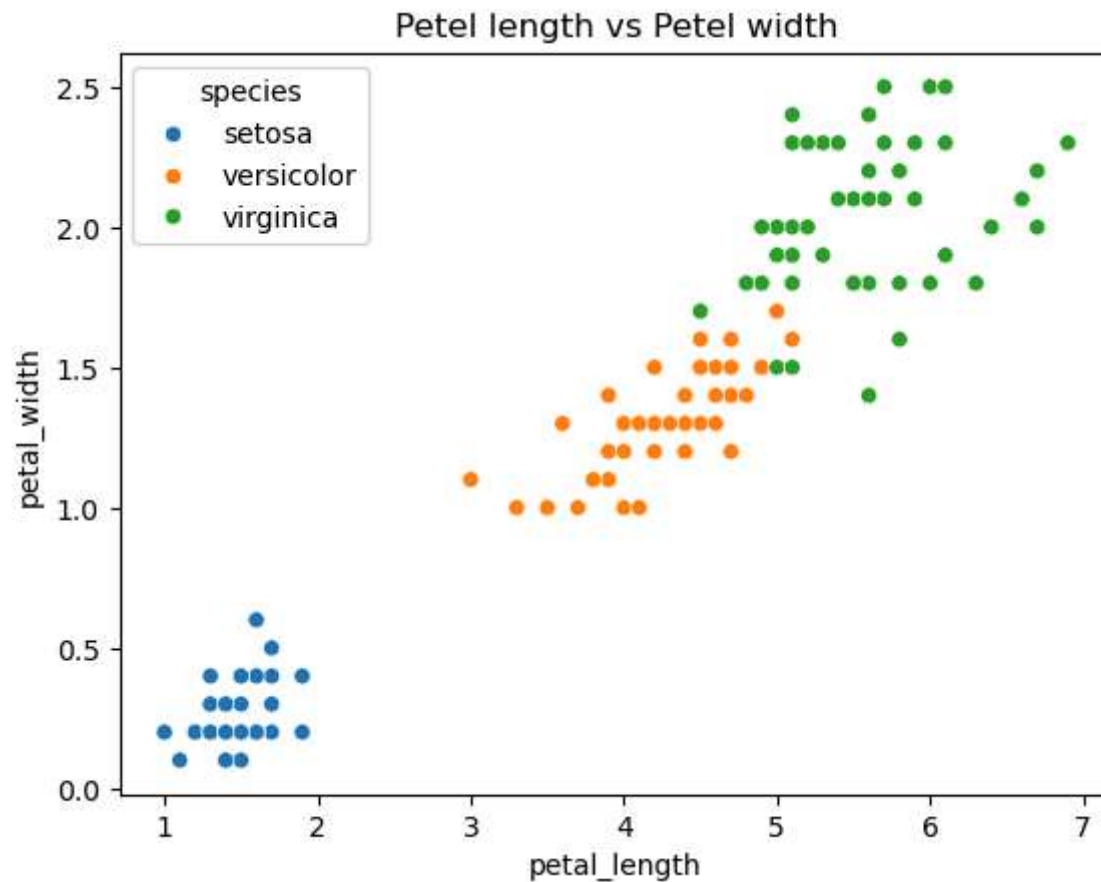
```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',  
      'species'],  
      dtype='object')
```

```
In [16]: print(df['species'].unique())
```

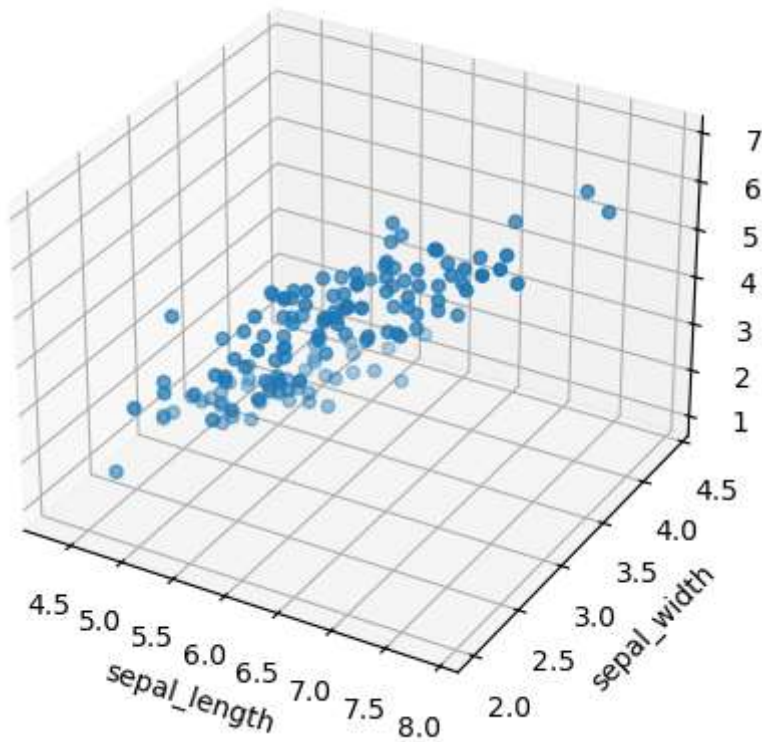
```
['setosa' 'versicolor' 'virginica']
```

```
In [17]: #Color Coded Scatter Plot
```

```
sns.scatterplot(  
    x='petal_length',  
    y='petal_width',  
    hue='species',  
    data=df)  
plt.title("Petel length vs Petel width")  
plt.show()
```



```
In [18]: #3D Scatter plot
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(8,5))
ax = fig.add_subplot(111,projection='3d')
ax.scatter(
    df['sepal_length'],
    df['sepal_width'],
    df['petal_length'])
ax.set_xlabel('sepal_length')
ax.set_ylabel('sepal_width')
ax.set_zlabel('petal_length')
plt.show()
```



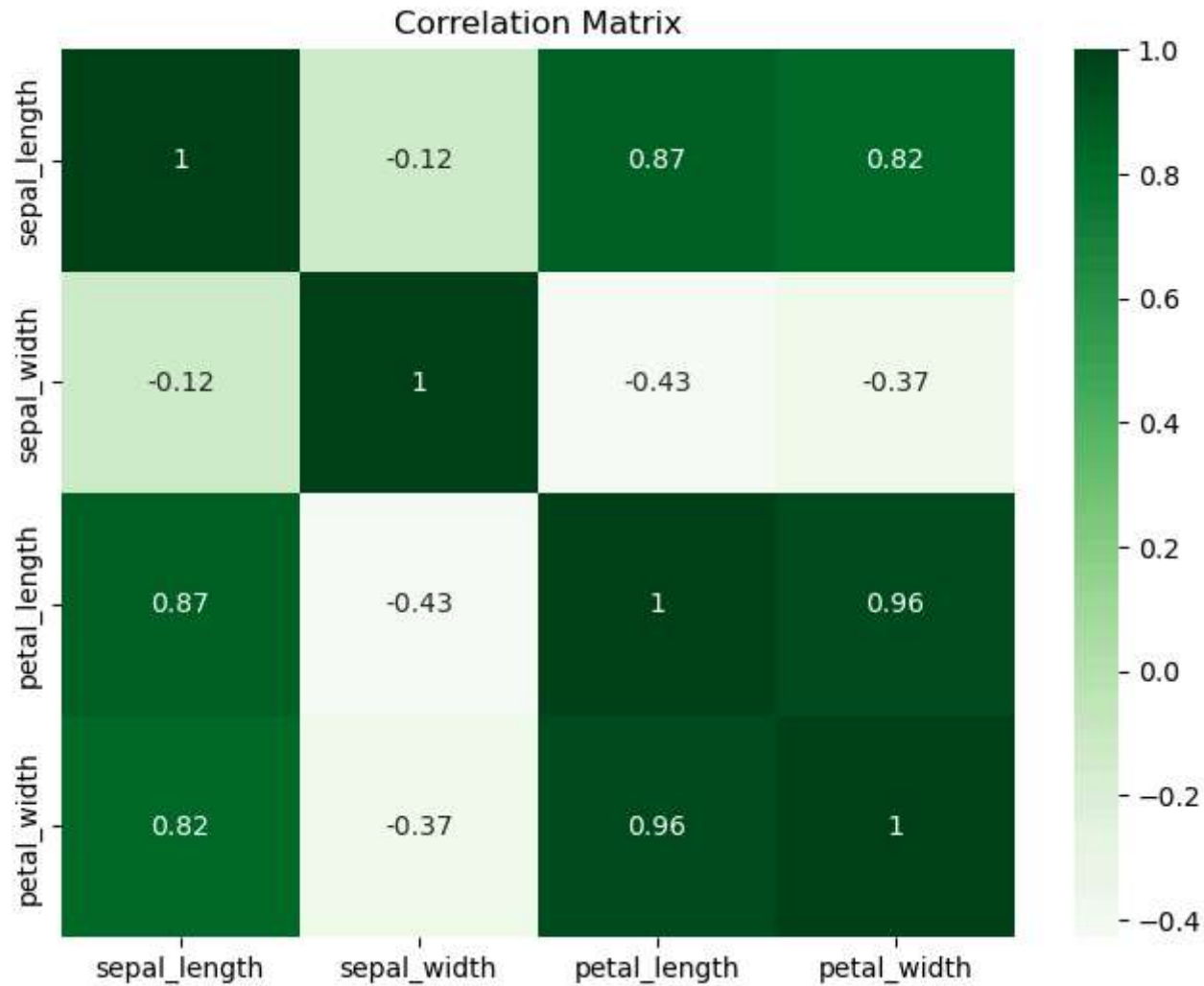
Correlation Analysis

```
In [19]: corr_matrix = df.corr(numeric_only=True)
print(corr_matrix)
```

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.117570	0.871754	0.817941
sepal_width	-0.117570	1.000000	-0.428440	-0.366126
petal_length	0.871754	-0.428440	1.000000	0.962865
petal_width	0.817941	-0.366126	0.962865	1.000000

```
In [20]: plt.figure(figsize=(8,6))
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='Greens'
```

```
)
plt.title("Correlation Matrix")
plt.show()
```



```
In [32]: #VIF - Variance Inflation Factor.
#1 - no correlation
#1-5 low correlation
#5-10 moderate
#>10 severe
from statsmodels.stats.outliers_influence import variance_inflation_factor
```

```
x=df.drop("species",axis=1)
vif=pd.DataFrame()
vif['Feature']=x.columns
vif['VIF']=[
    variance_inflation_factor(x.values,1)
    for i in range(x.shape[1])
]
print(vif)
```

	Feature	VIF
0	sepal_length	96.353292
1	sepal_width	96.353292
2	petal_length	96.353292
3	petal_width	96.353292

```
In [34]: #Remove a High VIF feature
#Suppose we remove petal_length.
X_new=x.drop("sepal_length",axis=1)
vif_after=pd.DataFrame()
vif_after["Feature"]=X_new.columns
vif_after["VIF"]=[
    variance_inflation_factor(X_new.values,1)
    for i in range(X_new.shape[1])
]
print(vif_after)
```

	Feature	VIF
0	sepal_width	62.071308
1	petal_length	62.071308
2	petal_width	62.071308

```
In [35]: #HEATMAP AFTER REMOVING FEATURE
plt.figure(figsize=(8,6))
sns.heatmap(
    X_new.corr(),annot=True,
    cmap="coolwarm"
)
plt.title("After VIF Treatement")
plt.show()
```